Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/26/2012
PI Name:	Turker, Mitchell Ph.D.		
Project Title:	The Relation Between Mutagenesis and Genomic Instability After Particle Exposure In Vivo		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Space Biology Special Category:	None		
PI Email:	turkerm@ohsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	503-494-2168
Organization Name:	Oregon Health & Science University		
PI Address 1:	3181 SW Sam Jackson Park Rd		
PI Address 2:	CROET/L606		
PI Web Page:			
City:	Portland	**State:**	OR
Zip Code:	97239-3011	**Congressional District:**	1
Comments:			
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Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kronenberg, Amy (Lawrence Berkeley National Laboratory)		
Grant/Contract No.:	NNX10AC12G		
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Task Description:	Highly energetic charged particles in space environments are hazardous because they cause DNA damage, which leads to mutation and loss of expression of critical genes that normally prevent cells from becoming cancerous. Most charged particle-induced mutations at autosomal loci span large regions of an affected chromosome resulting in loss of heterozygosity (LOH) for these regions. The proposed work has three interrelated goals. The first is to document the induction of mutational processes leading to LOH in two tissues, lung and spleen. The mutagenic effects of three charged particles will be tested: 1 GeV/amu Fe ions, 1 GeV/amu Ti ions, and 1 GeV protons to address the impact of ionization density on mutagenesis. The second goal is to demonstrate that LOH occurs on multiple chromosomes in the induced mutant cells, which would suggest the development of genomic instability. The third goal is to demonstrate directly with novel color markers that genomic instability characterized by high levels of LOH develops in the progeny of cells exposed in vivo. The basis for this approach is that selection of cells with LOH type mutations will provide the "bait" to		

	"fish out" cells exhibiting charged particle-induced genomic instability. Successful completion of the proposed work will identify those types of mutational events that reflect charged particle-induced genomic instability, and the role of ionization density in the development of these unstable cells in apparently normal tissues. The results will be useful in assessing cancer risks from space travel in two tissues that are prone to radiogenic cancers in humans.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Damage from ionizing radiation is both an environmental concern and a concern for cancers receiving therapies, most notable for cancer. Our work provides information on how ionizing radiation can damage cellular genomes.
Task Progress:	Overview. The objective of the ongoing project is to determine the effects of accelerated particles found in the space environment on cells in intact organs. These particles contain nuclei stripped of their electrons. The three particles under study are iron (Fe) ions, titanium (Ti) ions, and protons (a hydrogen nucleus). To determine the effects in the intact organs, live mice are exposed to the accelerated particles at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). The mice are then returned to Oregon Health Science University (OHSU) in Portland, OR for testing approximately 3-4 months after exposure at BNL. Radiation effects are measured in the cells from three tissues: lung, spleen, and kidney. We are specifically funded for the lung and spleen cells, but have added kidney for some assays because we have shown in over 8 years of prior work that the cells of this organ are sensitive to accelerated particles. Thus, the kidney serves as a good control that our assays are working and that the exposures at BNL were effective, and to continue to obtain data of particle effects on this tissue. Two end points are used. The first is to determine the extent of overall cellular harm (toxicity) from the particle exposure. The second is to determine the extent of harm to the cells' genetic material (i.e., DNA). Mice with a specific genetic background are bred for these studies. The work below will describe the results from our second and third set of experiments in which the mice were exposed to 25 and 50 cGy of Fe ions and 40 cGy of Ti ions, and additional work at higher doses. Data from protons are currently being examined Unexposed mice are used as controls. Cellular harm. The toxic effects from radiation exposure can be determined with a "cloning assay." For this assay, each lung or kidney is treated with enzymes to separate the cells, which are then plated in culture to determine how many cells can grow into a clone. A clone is the product of a single cell that can divide enough times to produce 100 or more progeny cells. The relative number of cells that can grow into clones provides the cloning efficiency for each organ. The spleen can be disrupted manually to create a single cell suspension for the same type of assay. Radiation effects are demonstrated by showing that the cloning efficiency is reduced for cells from organs exposed to radiation in the intact mouse compared with cells from the same organs obtained from non-irradiated mice. Cloning efficiencies were measured for lungs, kidneys, and spleens at 3-4 months after the mice were exposed to the lower doses of Fe ions and Ti ions. No toxicity was observed despite the DNA harm effects discussed below. DNA harm. DNA is the genetic material within each cell, and DNA damage has been linked to several disease processes including most notably cancer. DNA damage can cause mutations, which are small or large alterations in the DNA sequence. We use mice that have one good copy and one bad copy of a gene called Aprt to measure DNA harm. The important point about the Aprt mice for this discussion is that methods are available to separate cells that have lost the good Aprt copy from those that still retain the good copy. The cells that lose the good copy, which are called Aprt mutants, are present in the non-irradiated mice at frequencies of 1 in 100,000 cells for the spleen 1 in 10,000 cells for the kidneys and approximately 1 in 20-30,000 in the lungs. In other words the mutant cells are quite rare, but we have methods to find them. Radiation can increase the Aprt mutant frequency because it causes DNA damage, which occasionally causes loss of the Aprt gene. The Aprt mutant frequencies increased with statistical significance for the cells obtained from kidneys of mice exposed to 40 cGy Ti ions and 50 cGy Fe ions, but not for the lung or spleen cells. DNA harm can also be demonstrated by directly examining the DNA molecules from mutant cells for changes on the chromosome that contains the good copy of Aprt. We have obtained DNA from Aprt mutants for most of the conditions described above for this analysis and have shown mutagenic effects in both the kidneys and spleens exposed to 40 cGy Ti ions and 50 cGy Fe ions. Summary. We have successfully demonstrated cellular and DNA harm in the cells of living mice exposed to the types of accelerated particles found in the space environment. Most notably, we demonstrate effects at doses far lower than we have tested before and near those that astronauts could encounter in the space environment. During the next year we will examine even lower doses.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Turker M, Kronenberg A, Grygoryev D, Dan C, Ohrlich A, Eckelmann B, Gauny S. "Toxicity and Mutagenesis in Three Mouse Tissues Exposed to Accelerated Fe Ions, Ti Ions, or Protons." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Grossi G, Dan C, Grygoryev D, Turker M. "Autosomal mutations in murine kidney epithelial cells exposed to graded doses of densely ionizing 1 GeV/amu Fe ions." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Grossi G, Dan C, Grygoryev D, Turker M. "Autosomal mutations in murine kidney epitheial cells exposed to graded doses of densely ionizing 1 GeV/amu Fe ions." 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. 58th Annual Meeting of the Radiation Research Society, San Juan, Puerto Rico, September 30 – October 3, 2012. Presentation number: PS5-09. , Oct-2012